

Tom S. Bertalan

DATA SCIENTIST · BIOREACTOR CONTROLS · NEURAL NETWORKS FOR NONLINEAR DYNAMICS

Boston, MA

☎ (256) 613-3760 | ✉ Tom@TomBertalan.com | 🏠 TomSB.net | 📺 TSBertalan | 📄 Tom-Bertalan-00764640

*Data scientist specializing in automation, neural system identification,
and high-dimensional time-series data integration for real-time processing and optimization.*

PROFESSIONAL EXPERIENCE AND RESEARCH AREAS

Amgen

Cambridge, MA

DATA SCIENTIST

11/2024-Present

• Mechanistic Simulation & Model Qualification

- Author a significant release of an internal tool for mechanistic simulation of lyophilization with radiative secondary-drying physics; calibrate across multiple programs and vial types.
- Promulgate the Secondary Drying Report Card: a validation/qualification framework tracking model readiness.
- Design deployment-friendly objects and serialization; deliver containerized environments with reproducible notebooks.

• Cross-functional Delivery & Governance

- Lead cross-functional sprints and experimental design to accelerate predictive-model delivery.
- Create microlearning, demos, and onboarding materials to institutionalize model credibility and reproducible digital-twin practice.

University of Massachusetts

Lowell, MA

RESEARCH SOFTWARE ENGINEER

1/2024-11/2024

• Model Predictive Control of CHO Bioreactor

- Designed and administered virtualization, certs, remote access, and a Flask+Postgres+React bioprocess data lake.
- Integrated OPC-UA, MODBUS, and Java pump control subsystems for online distributed control of a 5L bioreactor.
- Collected targeted experimental data from a 3-liter bioreactor, and characterized dynamics and responses to control inputs.
- Supervised the implementation of more rigorous simulation methods for an SBML-encoded bioreactor digital twin.
- Developed an optimal online model-predictive controller in Aspen DMC to maximize economic factors.

Johns Hopkins University

Baltimore, MD

POSTDOCTORAL FELLOW

2/2020-1/2024

• Machine Learning Service Deployment and Administration

- Provisioned local Llama2 inference server over VPN for zero-shot microservices (doc renaming, topic extraction, categorization).
- Automated conversion of seminar announcements to ICS files via OpenAI API, including curation of a 19.7k-word fine-tuning dataset.
- Wrote specifications and solicited bids for an multi-GPU training server destined for datacenter colocation.

• Neural Differential Equations and Time-Series

- Advanced SotA for neural DEs for time series, including CNNs for PDEs; conceived a novel loss function for Hamiltonian systems.
- Cut RNN inference burn-in from 25 to 5 samples using manifold learning.
- Derived custom gradients for nets with constraints and promoted new theory on neural ODE error scaling laws.
- Parallelized neural PDE solution sensitivity analysis using SLURM on GPU cluster.

• Modeling and Experimental Design for CHO Metabolism

- Worked cross-functionally as a theorist/software engineer with experimental biologists on CHO hybrid mechanistic/ML metabolic modeling, leading the development of Python/MATLAB tools for simulation and data processing.
- Created a GUI for Bayesian experimental design; mentored team members on its use and maintenance.

• Robotic Perception and Control

- Developed a variational autoencoder for end-to-end robotic localization.
- Trained a U-net on both open and custom datasets for real-time (>10hz) on-board semantic segmentation of drivable space.
- Inferred depth and semantics in simulation using pre-trained transformers with GPU-accelerated real-time inference.

The Massachusetts Institute of Technology

Cambridge, MA

POSTDOCTORAL ASSOCIATE

3/2018-2/2020

• Autonomous Vehicle Design and Pathfinding

- Developed a model AV with firmware-level speed sensing and control commanded by checksummed bus communications.
- Designed a jerk-minimizing path planner capable of planing up to two lane switches ahead at 47 mph.
- Leveraged Ipopt and CppAD to push a model-predictive path follower to a latency of 67 ms in simulation.
- Detected dashcam obstacles via windowed support-vector machine.

• Nonlinear dynamics in neuroscience

- Wrote object-oriented library for fine- and coarse-grained simulation of neuronal dynamics.
- Analyzed bifurcation and resonance behavior of a mammalian circadian rhythm model.
- Coauthored successful application for a \$1.8M industry-academic partnership grant.

Built a rover for particle-filter SLAM with LIDAR using Gazebo and ROS, and a custom library for visualizing OpenCV pipelines and dynamic pruning of computational graphs. Simulated thousands of neurons in Numpy and MATLAB, and social agents in OpenMP-accelerated C++.

SKILLS

Libraries, frameworks, & tools

- Python/C++/Java
- NumPy+SciPy+Matplotlib
- Pytorch/TensorFlow/Keras
- Linux, shell scripting, and git
- VSCode/Eclipse+PyDev
- Aspen DMC3 Controls
- OPC-UA, MODBUS/serial, and ROS

Areas of Expertise and Training

- Nonlinear dynamics of time series data
- Deep learning and ML
- Dimension reduction and representation learning
- Probabilistic modeling and theory
- Computer vision
- Research presentation and dissemination
- Trainee and peer mentoring

Other Skills and Interests

- Home automation with Arduino, Raspberry Pi, and 3D printing
- Solo and orchestral violin performance
- Windsurfing and small-boat sailing

EDUCATION & TRAINING

Institutions

- **Johns Hopkins University** **2020 - 2024**
Postdoctoral Fellow *Che. & Bio. Eng.*
- **The Massachusetts Institute of Technology** **2018 - 2020**
Postdoctoral Associate *Mech. Engr.*
- **Princeton University** **2012 - 2018**
NSF Research Fellow *PhD & MA, Che. & Bio. Eng.*
- **The University of Alabama** **2008 - 2012**
Student Research Assistant *BS Che. & Bio. Eng.; Minor in Math*

Awards

- Princeton Program in Plasma Science and Technology research and academic fellowship
- National Science Foundation research fellowship
- William R. Schowalter fund for scholarly conferences
- National merit finalist scholarship
- University honors program and president's list
- $\Phi\chi\Sigma$, $\Omega\chi E$, and $TBII$ honor societies

Teaching

- Taught a recurring summer course for freshmen on employing ROS, OpenCV, and CNNs for lane det. and traffic sign recog.
- Supervised operations and measurement activities in junior Chemical & Biological Engineering practicum.
- Lectured, held office hours, and graded assignments in senior differential equations course.

SELECTED PUBLICATIONS

Integrated Modeling and Optim. of Secondary Drying for Pharma. Freeze-Drying Proc.

Tom Bertalan, M. Maritato, A. Ganguly, W. Al Bakri, R. Tukra, B. Lacetti

2025 (accepted)
AIChE 2025 (poster)

Integrated Continuous USP Platform for Max. Productivity and Closed-Loop Controlled CQA

J. Lee, Tom Bertalan, Z. Wang, Y. Choi, S. Song, H. Yamanaka, A. Rajendran, J. Han, S. Yoon

2025 (accepted)
AIChE 2025

Digital Twin-Enabled Control of Productivity and CQA in CHO Cell Cultures

Y. Choi, S. Yoon, Z. Wang, Tom Bertalan, J. Lee

2025 (accepted)
AIChE 2025 (poster)

Beyond Trial-and-Error: An in silico Model for Accelerated Peristaltic Fill Recipe Development

Maritato, Ingram, Sao, Heymann, Tom Bertalan, Banerjee, Icten Gencer, Ganguly, Pearson, Schlegel

2025 (accepted)
AIChE 2025 (poster)

Integ. of Bayesian Optim. and Solution Thermo. to Opt. Media Design for Mammalian Biomfg.

Nelson Ndahiro, Edward Ma, Tom Bertalan, Marc Donohue, Yannis Kevrekidis, and Michael Betenbaugh

2025
iScience

Data-driven and Physics Informed Modelling of Chinese Hamster Ovary (CHO) Cell Bioreactors

T. Cui, Tom Bertalan, N. Ndahiro, P. Khare, M. Betenbaugh, C. Maranas, I. Kevrekidis

2024
Comp. & Chem. Engr.

Symbolic regression and modular neural differential equations for bioprocess engineering and robotics

Tom Bertalan, Jaeweon Lee, Zhao Wang, Seongkyu Yoon, I. Kevrekidis

2024
MLDS 4 (poster)

Learning emergent PDEs in a learned emergent space

F. Kemeth, Tom Bertalan, T. Thiem, S. Moon, C. Laing, I. Kevrekidis

2022
Nature Comm.

Initializing LSTM internal states via manifold learning

F. Kemeth, Tom Bertalan, N. Evangelou, T. Cui, S. Malani, I. Kevrekidis

2021
Chaos

PUBLICATIONS

Integ. of Bayesian Optim. and Solution Thermo. to Opt. Media Design for Mammalian Biomfg. Nelson Ndahiro, Edward Ma, Tom Bertalan , Marc Donohue, Yannis Kevrekidis, and Michael Betenbaugh	2025 <i>iScience</i>
Machine Learning Approaches to Problem Well-Posedness Tom Bertalan , G. Kevrekidis, E. Rebrova, S. Mishra, I. Kevrekidis	<i>In Preparation</i>
Data-driven and Physics Informed Modelling of Chinese Hamster Ovary (CHO) Cell Bioreactors T. Cui, Tom Bertalan , N. Ndahiro, P. Khare, M. Betenbaugh, C. Maranas, I. Kevrekidis	2024 <i>Comp. & Chem. Engr.</i>
Implementation and (Inverse Modified) Error Analysis for implicitly-templated ODE nets A. Zhu, B. Zhu, Tom Bertalan , Y. Tang, I. Kevrekidis	2024 <i>SIAMDS</i>
Transformations establishing equivalence across neural networks: when have two networks learned the same task? Tom Bertalan , F. Dietrich, I. Kevrekidis	2024 <i>Chaos</i>
Certified Invertibility in Neural Networks via Mixed-Integer Programming T. Cui, Tom Bertalan , G. Pappas, M. Morari, I. Kevrekidis, M. Fazlyab	2023 <i>L4DC 2023 — PMLR</i>
Some of the variables, some of the parameters, some of the times, with some things known: Identification with partial information Saurabh Malani, Tom Bertalan , T. Cui, M. Betenbaugh, J. Avalos, I. Kevrekidis	2023 <i>Comp. & Chem. Engr.</i>
Learning effective stochastic differential equations from microscopic simulations: linking stochastic numerics to deep learning F. Dietrich, A. Makeev, G. Kevrekidis, N. Evangelou, Tom Bertalan , S. Reich, I. Kevrekidis	2023 <i>Chaos</i>
Learning emergent PDEs in a learned emergent space F. Kemeth, Tom Bertalan , T. Thiem, S. Moon, C. Laing, I. Kevrekidis	2022 <i>Nature Comm.</i>
Personalized Algorithm Generation: A Case Study in Meta-Learning ODE Integrators Y. Guo, F. Dietrich, Tom Bertalan , D. Doncevic, M. Dahmen, I. Kevrekidis, Q. Li	2022 <i>SIAM J. Sci. Comp.</i>
Initializing LSTM internal states via manifold learning F. Kemeth, Tom Bertalan , N. Evangelou, T. Cui, S. Malani, I. Kevrekidis	2021 <i>Chaos</i>
Development of closures for coarse-scale modeling of multiphase and free surface flows using machine learning C. Linares, Tom Bertalan , E. Koronaki, Jicai Lu, G. Tryggvason, I. Kevrekidis	2021 <i>APS Bulletin</i>
Global and local reduced models for interacting, heterogeneous agents T. Thiem, F. Kemeth, Tom Bertalan , Carlo Liang, I. Kevrekidis	2021 <i>Chaos</i>
Local conformal autoencoder for standardized data coordinates E. Peterfreund, O. Lindenbaum, F. Dietrich, Tom Bertalan , M. Gavish, I. Kevrekidis, R. Coifman	2020 <i>PNAS</i>
Emergent spaces for coupled oscillators T. Thiem, M. Kooshkbaghi, Tom Bertalan , Carol Laing, I. Kevrekidis	2020 <i>Front. in Comp. Neuro.</i>
On Learning Hamiltonian Systems from Data Tom Bertalan , F. Dietrich, Igor Mezic, and I. Kevrekidis	2019 <i>Chaos</i>
An Emergent Space for Distributed Data with Hidden Internal Order through Manifold Learning F. Kemeth, Sindre Haugland, F. Dietrich, Tom Bertalan , Kevin Höhle, Q. Li, Erik Bollt, Ronen Talmon, Katharina Krischer, and I. Kevrekidis	2017 <i>IEEE Access</i>
Coarse-grained descriptions of dynamics for networks with both intrinsic and structural heterogeneities Tom Bertalan , W. Wu, C. Laing, C. William Gear, and I. Kevrekidis.	2017 <i>Front. in Comp. Neuro.</i>
Dimension reduction in heterogeneous neural networks: Generalized Polynomial Chaos (gPC) and ANALYSIS-Of-VARiance (ANOVA) M. Choi, Tom Bertalan , C. Laing, and I. Kevrekidis.	2016 <i>Euro. Phys. J., Special Topics</i>
OpenMG: a new multigrid implementation in Python Tom Bertalan , A. Islam, R. Sidje, and E. Carlson	2014 <i>Num. Lin. Alg. with App.</i>

PRESENTATIONS

Integrated Modeling and Optim. of Secondary Drying for Pharma. Freeze-Drying Proc.

Tom Bertalan, M. Maritato, A. Ganguly, W. Al Bakri, R. Tukra, B. Lacetti

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2025 (accepted)

AIChE 2025

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Symbolic regression and modular neural differential equations for bioprocess engineering and robotics

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2024

MLDS 4 (poster)

Certified Invertibility in Neural Networks via Mixed-Integer Programming

T. Cui, Tom Bertalan, G. Pappas, M. Morari, I. Kevrekidis, M. Fazlyab

2023

Learning for Dyn. Sys.

Coarse-grained and emergent distributed-parameter systems from data

Hassan Arbabi, F. Kemeth, Tom Bertalan, I. Kevrekidis

2021

American Control Conf.

Data-driven model reduction and discovery

T. Thiem, Tom Bertalan, F. Kemeth, Yorgos Psarellis, I. Kevrekidis

2020

AIChE

Dynamical-systems-guided learning of PDEs from data

Hassan Arbabi, Tom Bertalan, A. Roberts, I. Kevrekidis

2020

AIChE

On the data-driven discovery and calibration of closures

S. Lee, Yorgos Psarellis, C. Siettos, Tom Bertalan, D. Amchin, T. Bhattacharjee, S. Datta, I. Kevrekidis

2020

AIChE

Connections between residual networks and explicit numerical integrators, and applications to identification of noninvertible dynamical systems

T. Cui, Tom Bertalan, Yorgos Psarellis, I. Kevrekidis

2020

AIChE

Neural network approach to reduced order modeling of multiphase flows

C. Linares, Tom Bertalan, S. Lee, J. Lu, G. Tryggvason, I. Kevrekidis

2020

APS Div. of Fluid Dyn.

PDE+PINN: Learning and Solving a PDE at the Same Time

Tom Bertalan, F. Kemeth, T. Cui, I. Kevrekidis

2020

AIChE

Learning Partial Differential Equations from Discrete Space Time Data: Convolutional and Recurrent Networks, and Their Relations to Traditional Numerical Methods

Tom Bertalan, F. Dietrich, T. Thiem, R. Farber, I. Kevrekidis, A. Roberts

2019

AIChE

Recurrent Neural Networks, Numerical Integrators and Nonlinear System Identification

Tom Bertalan, R. Farber, T. Thiem, F. Dietrich, I. Kevrekidis

2018

AIChE

Coarse-Scale PDEs from Microscopic Observations Via Machine Learning

S. Lee, M. Kooshkbaghi, C. Siettos, Tom Bertalan, and I. Kevrekidis

2019

AIChE

When Have Two Networks Learned the Same Task? Data-Driven Transformations between System Representations

Tom Bertalan, F. Dietrich, T. Thiem, I. Kevrekidis

2019

AIChE

Coarse modeling of circadian rhythms in heterogeneous neural networks

Tom Bertalan, C. William Gear, Yannis G. Kevrekidis, M. Henson, E. Herzog, and C. Laing

3017; 2016

Dyn. Days 2017; AIChE

Coarse-graining of heterogeneous neural dynamics

Tom Bertalan, M. Choi, C. Laing, I. Kevrekidis

2015

AIChE

Heterogeneity and reduction for complex network dynamics

I. Kevrekidis, A. Holiday, Tom Bertalan, and C. Laing

2014

AIChE

Coarse-graining Network Dynamics

A. Holiday and Tom Bertalan

2013

Network Front.

nSpyres: an open-source, Python-based framework for simulation of flow through porous media

E. Carlson, A. Islam, F. Dumkwu, and Tom Bertalan

2012

Interpore 2012

OpenMG: a new multigrid implementation in Python

Tom Bertalan, A. Islam, R. Sidje, and E. Carlson

2012

Proc. 11th Python in Sci. Conf.

ESIM: a framework for simulation of dominance hierarchy formation in small animal groups

Tom Bertalan and R. Earley

2012

UA Hon. Undergr. Res. Conf.

An open-source computing cluster for virtual experiments with variable parameters

Tom Bertalan and E. Carlson

2011

UA Hon. Undergr. Res. Conf.